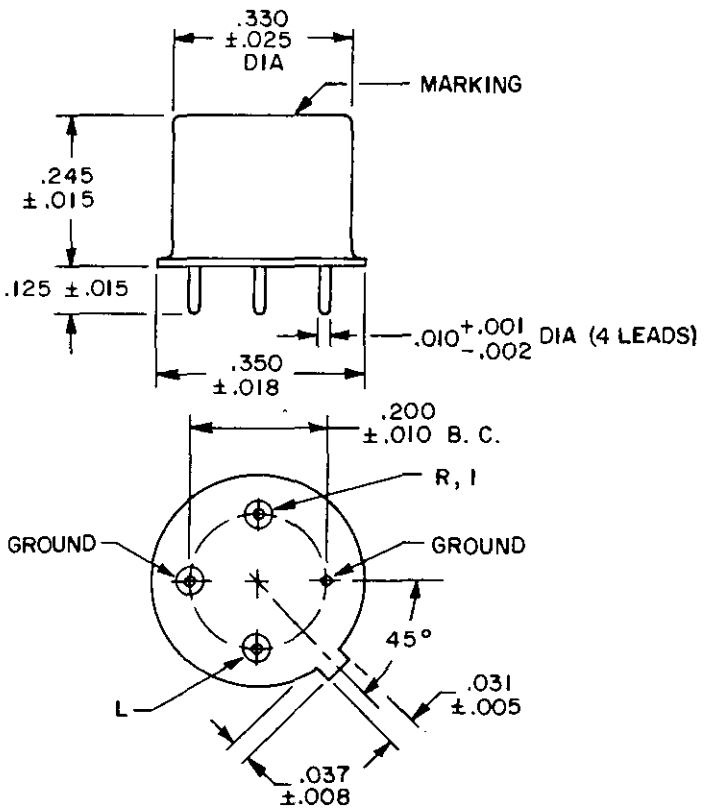
MILITARY SPECIFICATION SHEET

MIXER STAGES, RADIO FREQUENCY, SINGLE BALANCED, PC MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring mixers described herein shall consist of this document and the latest issue of Specification MIL-M-28837.



INCHES	MM
.001	.03
.002	.05
.005	.13
.008	.20
.010	.25
.015	.38
.018	.46
.025	.64
.031	.79
.037	.94
.125	3.18
.200	5.08
.245	6.22
.330	8.38
.350	8.89

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Outline drawing for mixers P/N M28837/8-01N and 01S.

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FSC 5985

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.

Electrical characteristics:
Operating frequency range: RF and IF: DC - 800 MHz; LO: 5 - 800 MHz.
LO drive power: +7 dBm nominal (+7 dBm used as the test value for the electrical requirements).
Conversion loss (max): f_L and f_R 5-500 MHz, f_I DC-500 MHz = 6.5 dB

f_L and f_R 500-800 MHz, f_I DC-800 MHz = 7.0 dB.
Noise figure (SSB): f_L and f_R 5-500 MHz f_I 0.4-500 MHz = 6.5 dB

f_L and f_R 500-800 MHz f_I 0.4-800 MHz = 7.0 dB

Isolation (minimum):	LO-RF	LO-IF	LO frequency
	40	40	5-50 MHz.
	30	30	50-100 MHz
	20	20	100-800 MHz.

Maximum input power: 18 mW rms.

DC polarity: Negative with two in-phase signals applied to the LO and RF ports.

Physical and Environmental characteristics:
Weight: 1.3 grams.
Temperature: Operating: -65° to +100°C.
Nonoperating: -65° to +100°C.
Shock: Test condition C.
Seal: Hermetic.
Terminal strength: Test condition C.

Part number M28837/8-01N, 01S.

NOTE: Mixers covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	FSN	Manufacturer's code	Commercial type
01N	---	14482	M7C
01S	---	97942	581R869H01
01S	---	14482	M7C

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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